# Project Report

02180 - Introduction to Artificial Intelligence
Technical University of Denmark

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## Group number: 24

## Group Composition


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# 1 Game rules

*Game rules, in particular if you choose a game not on the list above or if you make simplifications to the game. Make this part as short as possible. If you have simplified the rules, explain why and explain what it would take to make AI for the full game.*

# 2 Game Type

*What kind of game is it? Relevant questions are e.g.: Single- or multi-player? Competitive or cooperative? Zero-sum? Is it turn-based, concurrent or real-time? Perfect or imperfect information (full or partial observability)? Deterministic or stochastic? And what does this imply for the kind of algorithms that apply to the game?*

# 3 Size space

*What, approximately, is the size of the state space of the game, that is, the number of reachable states from an initial state of the game? Finding the precise number can often be very difficult, but here it is enough to estimate whether it is e.g. in the ballpark of* $10^5$, $10^{10}$, $10^{15}$, $10^{20}$*... You can e.g. try to give an upper bound on the size of the state space (or both a lower and upper bound). What does this imply for the kind of algorithms that apply to the game?*

# 4 Elements of the game

*Describe the following elements of your game (or the relevant subset):* $s_0$ *(initial state), Player(s), Action(s), Result(s, a), Terminal-Test(s), Utility(s, p). You don't necessarily have to specify these formally, but try to describe them as clearly and precisely as possible. If your chosen game cannot be expressed fully in terms of these functions, explain why, and describe the additional components as precisely as possible.*

# 5 AI states and moves representation

*How are states and moves represented in the computer? To make AI for this game, was it enough to represent a game state as the position of the pieces or the distribution of the cards? Or was it e.g. necessary to represent game states as belief states, and why? Don't give too implementation-specific answers, but answer in overall terms of algorithms and data structures.*

# 6 Methods & Algorithms

*Which of the methods and algorithms considered in the course could potentially be used to construct AI for your game, and which have you chosen? If you use one of the standard algorithms without modification, you don't have to describe it, but can just refer to the textbook (or other sources). However, try to give a small concrete example of how the algorithm works in your game, e.g. by illustrating a few steps of the search graph/tree* *being built (you can do this in a smaller or simplified version of the game, if you prefer). Also, argue of the appropriateness of your chosen algorithm for the chosen game.*

# 7 Heuristics and Evaluation functions

*If you use heuristics or evaluation functions, make sure to explain how these have been designed. Use mathematical notation rather than implementation-specific notation.*

# 8 Implemented AI's parameters

*Are there any parameters that can be adjusted in your implemented AI? Did you try out several different algorithms, search depths, heuristics or evaluation functions? If so, it is a very good idea to include benchmarks (test results) comparing the different versions. It is also interesting to compare the strength of the AIs to human players (yourselves).*

# 9 Future work

*Comments on future work: How could your solution be improved? Could it be optimised using the same algorithms and data structures that you are currently using? Or would significant improvements of the AI require different methods, and if so, which could be considered?*